

Unit 13, Lesson 1: Effects of Outliers

Benchmark

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.

Learning Target

- ☐ I can determine the implications an outlier has in a given data set.

13.1.1 Warm-Up: Precision Is Key

Becky Miller is a quality control engineer with GE Aviation. Like many other careers, manufacturing engineering has many moving parts that make up an entire process. Becky says that manufacturing is problem-solving.

1. List at least two steps that you take when faced with a difficult task.

Becky mentions that an error “about the size of a human hair” can make a difference in manufacturing jet engines.

2. What is a time that you have made a small error that caused a big impact in your next steps?

13.1.2 Refresher: Measures of Center and Variation



Measures of center, such as **mean** and **median**, are used to describe a typical value in the data set.



Measures of variation, such as **range** and **interquartile range**, are used to describe how “spread out” the data is.

1. Fill in the blanks with the terms provided.

mean

range

median

interquartile range(IQR)



The _____ is the arithmetic average of a set of numbers. It is a measure of center, or central tendency.



The _____ is the difference between the highest data value and the lowest data value in a data set. It is a measure of variation.



A measure of variation in a set of numerical data, the _____ is the distance between the first and third quartiles of the data set.



The _____ is the middle of an ordered list of the values. If the list has an odd number of values, it is the middle value of that list. If the list has an even number of values, it is the average of the two middle values. It is a measure of center, or central tendency.

Suppose an office tracks the number of phone calls received on business days, Monday through Friday, over two weeks. The data set is shown below.

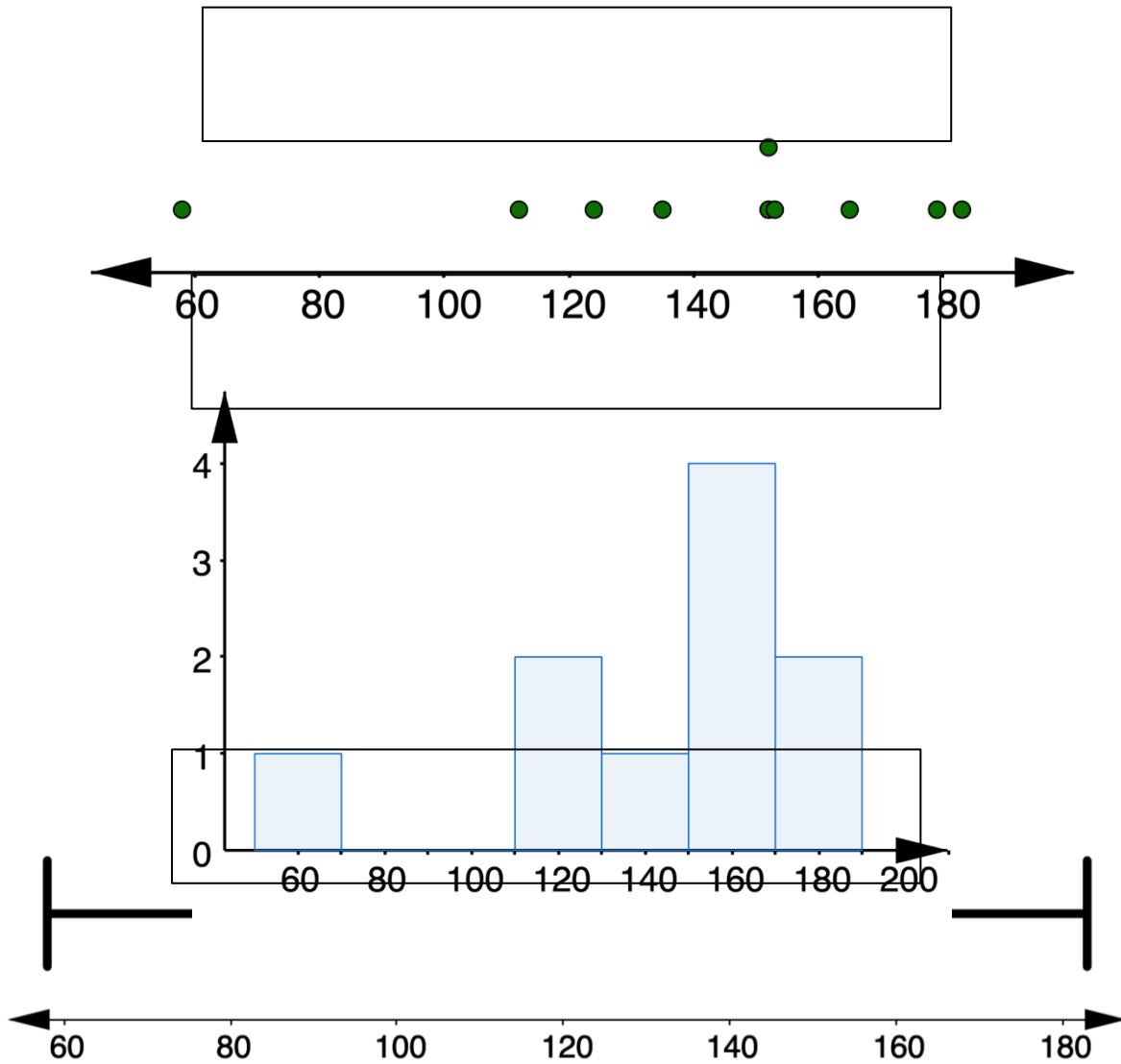
183	165	58	152	179
152	153	135	124	112

2. Discuss with your partner what you notice about the data set.
3. Match the measures of center and variation with their calculations for the data set above.

A. Mean	B. Median	C. Range	D. IQR
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	$\frac{152+152}{2} = 152$
	$165 - 124 = 41$
	$\frac{183 + 165 + 58 + 152 + 179 + 152 + 153 + 135 + 124 + 112}{10} = 141.3$
	$183 - 58 = 125$

4. A **histogram**, **line plot**, and **box plot** representing the data are shown.



- a. Label the display types by filling in the box above each display.

- b. Complete the table by identifying which measures of center and variation can be determined from each display.

	Mean	Median	Range	IQR
Histogram	☐	☐	☐	☐
Line Plot	☐	☐	☐	☐
Box Plot	☐	☐	☐	☐

5. Are there any data values that are far away from the rest?



An **outlier** is a value that is much higher or much lower than the other values in a set of data.

13.1.3 Guided Instruction: Stem-and-Leaf Plots

Recall the data collected on the number of phone calls an office received on business days over two weeks.

183	165	58	152	179
152	153	135	124	112

In addition to displaying the data graphically with a histogram, line plot, or box plot, the data can also be displayed in a ***stem-and-leaf*** plot.



A ***stem-and-leaf plot*** is a table that organizes data by place value to compare data frequencies.

- What do you notice about the data in the stem-and-leaf plot?

Stem	Leaf
5	8
6	
7	
8	
9	
10	
11	2
12	4
13	5
14	
15	2, 2, 3
16	5
17	9
18	3

Key: 5|8 = 58

- Compare the stem-and-leaf plot with the line plot from the refresher.

What similarities do you notice between the displays?	What differences do you notice between the displays?

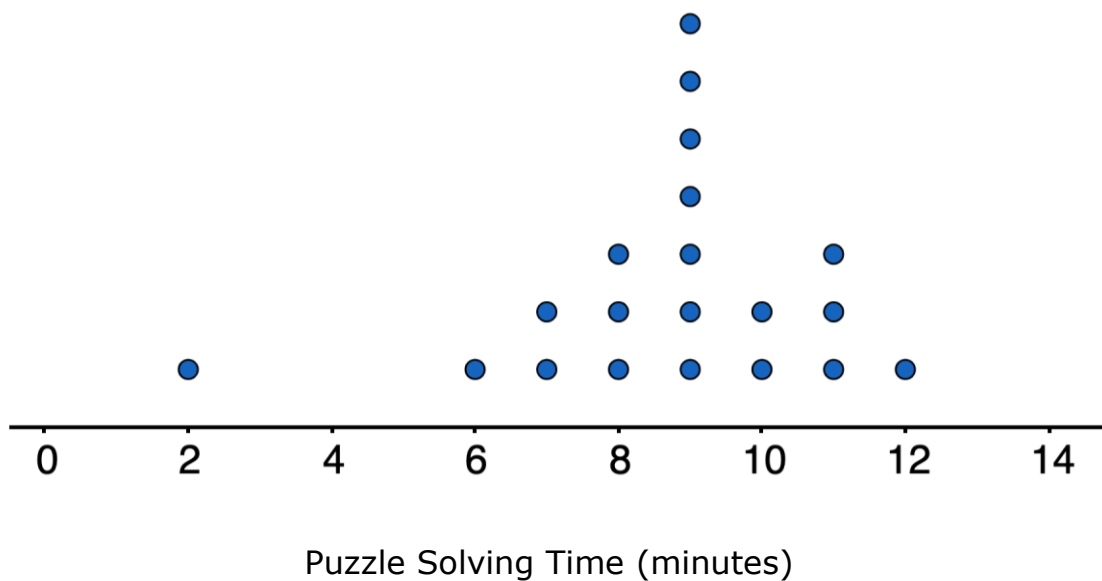
3. What value does the entry 13|5 represent?
4. Which measure(s) of center can be determined from the stem-and-leaf plot?
- ☐ mean
 - ☐ median
 - ☐ both the mean and median
5. Which measure(s) of variation can be determined from the stem-and-leaf plot?
- ☐ range
 - ☐ interquartile range
 - ☐ both the range and the interquartile range

13.1.4 Collaboration: Determining, Describing, and Comparing Numerical Data Represented Graphically

Work with your partner to complete the following.

1. Twenty students timed how long it took each of them to solve a puzzle. Below is a list of their solution times, in minutes, and a line plot displaying the data.

2, 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 9, 9, 9, 10, 10, 11, 11, 11, 12



- a. Determine the mean of the data set. Then, draw an arrow ($\uparrow$) below the number line that points to this value.
- b. Determine the median of the data set. Then, circle the value on the number line that represents the median.

- c. Discuss with your partner what you notice about the mean and the median of the data. Summarize your discussion.
- d. What is the range of the data?
- e. Find the interquartile range (IQR).

Q1	Q3	IQR

2. Imagine that the fastest time of 2 minutes was removed from the data set. Discuss with your partner what effect, if any, removing this value has on the values of center and variation (mean, median, range, IQR). Summarize your discussion.
3. Describe how the shape of the graph would change if 2 is removed.

4. Complete the table to compare the values of each measure before and after 2 is removed.

	Original Value	$>, <, =$	New Value After 2 is Removed
Mean			
Median			
Range			
IQR			

5. Which measure(s) of center will be affected if the outlier is removed?

- ☐ mean
- ☐ median
- ☐ both the mean and median

6. Which measure(s) of variation will be affected if the outlier is removed?

- ☐ range
- ☐ interquartile range
- ☐ both the range and the interquartile range

13.1.5 Your Turn!

The following data set shows the number of tickets a movie theater sold in one day for each movie that is playing at the theater.

162	623	76	277	181	64	105
265	82	263	124	167	150	245

1. One of the values in the data set is an outlier. Explain which value you think is the outlier.

2. Complete the statement.

Removing the outlier, 623, causes the mean to

- ☐ increase
- ☐ decrease
- ☐ remain the same

and the range to

- ☐ increase
- ☐ decrease
- ☐ remain the same

because it is much

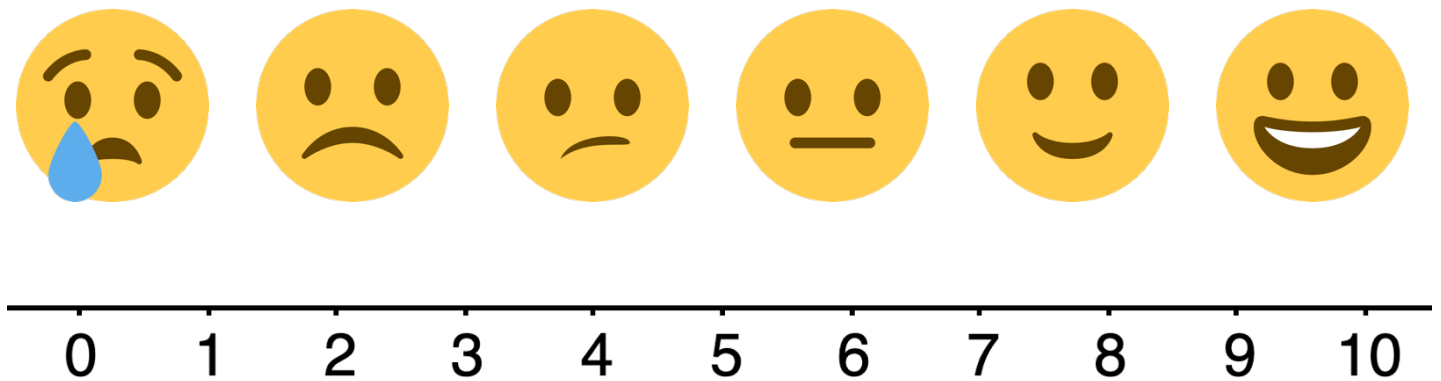
- ☐ larger
- ☐ smaller

than the other values in the data set.

13.1.6 Wrap-Up: Reflection

1. Rate yourself on today's learning target using the scale below.

I can determine the implications an outlier has in a given data set.



2. Write an explanation of why you rated yourself where you did.

